

# Measuring Inequality After Taxes and Transfers: A Reproducible Distributional Panel for the United States, 1979–2022

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Independent pre-doctoral research project

Source code and data: [github.com/Patrickkp1/tax-transfer-inequality](https://github.com/Patrickkp1/tax-transfer-inequality)

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## Abstract

The U.S. Census Bureau’s official income measure, used in nearly all public discussion of inequality, was designed in 1947 around an economy in which roughly 90% of household compensation arrived as cash. That economy no longer exists: by 2017 about two-thirds of federal transfer payments to households were non-cash (Medicare, Medicaid, SNAP, housing assistance, the EITC, the refundable Child Tax Credit), and the federal tax code had become much more progressive than it was in the early postwar period. Census’s pretax, cash-only construct counts neither side of the fiscal system, and the gap between that measure and a complete income concept has grown substantially over time. This paper builds a reproducible distributional accounting panel covering all five income quintiles plus eight upper-tail groups (81–90, 91–95, 96–99, Top 1, Top 0.1, Top 0.01, Top 0.001%, and the Forbes 400) for tax years 1979–2022, supplementing Current Population Survey (CPS) microdata with control totals from the Congressional Budget Office, Internal Revenue Service, Office of Management and Budget, Bureau of Economic Analysis, Bureau of Labor Statistics, Social Security Administration, and Giving USA. Once taxes and transfers are accounted for, the top-to-bottom quintile income ratio in 2022 is  $4.4\times$ , not the  $96\times$  gap implied by earned income alone, and real after-tax-and-transfer income has risen for every quintile of the U.S. distribution since 1979.

## 1 Motivation

In *The Myth of American Inequality* (2022), Phil Gramm, Robert Ekelund, and John Early argue that the standard Census Bureau measure of household income provides a distorted picture of the U.S. income distribution because it omits two large categories of income flows. First, the Census measure counts only cash transfers as income; it excludes Medicare, Medicaid, SNAP, housing assistance, and the refundable portions of the EITC and Child Tax Credit — in 1947 these represented less than 10% of all transfers, but by 2017 they were roughly two-thirds.<sup>1</sup> Second, the Census measure is a pretax measure: it ranks households by income before federal, state, and local taxes are deducted, even though households at the top of the distribution lose about 33% of pretax income to taxes and households at the bottom lose about 7%. Both omissions push the official inequality measure upward, and both have grown over time.

This project replicates Gramm et al.’s data architecture end-to-end from public sources, with two main motivations:

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<sup>1</sup>Gramm, Ekelund, and Early, *The Myth of American Inequality* (Rowman & Littlefield, 2022), pp. 13–15.

1. To produce a fully reproducible panel that any researcher can rebuild from the original administrative documents, with every assumption traceable to a specific data file or peer-reviewed citation. The seven-stage pipeline described below ingests CPS microdata, eleven CBO supplemental tables, IRS Statistics of Income files, OMB Historical Tables, BEA NIPA tables, BLS expenditure surveys, and SSA trust fund data.
2. To examine how the gap between Census’s measure and a complete post-fiscal measure has evolved over a long horizon (1979–2022), and what that gap implies for the standard narrative about rising inequality.

## 2 Conceptual framework

The income concept we attempt to measure for each household is

$$IAT = E + G + P - T \tag{1}$$

(where IAT denotes “income after taxes and transfers”),

where  $E$  is earned income (wages plus employer benefits plus business and financial income),  $G$  is government transfers (the cash and non-cash sum),  $P$  is private transfers (charitable transfers from households and institutions to households), and  $T$  is the household’s total federal, state, and local tax burden. This is the conventional “post-fisc” concept used in the CBO’s distributional analyses; we extend the CBO framework backward to 1979 and forward to 2022, and we add private transfers as a fourth component.

The Census “money income” measure used in headline inequality statistics is conceptually  $E + G_{cash}$  — it omits in-kind transfers ( $G - G_{cash}$ ), all of  $P$ , and all of  $T$ . Households are also ranked by money income, not by a post-fisc concept, so a low-income retiree receiving \$30k in Medicare benefits and a working-age household earning \$30k from wages appear in the same Census quintile despite having very different consumption capacity.

## 3 Data and method

The pipeline produces five panel CSVs in five sequential stages, plus a final ultra-top stage that reconciles the quintile-level panels with administrative data on the very wealthy. All scripts are written in Python and are designed to be self-contained and re-runnable from the underlying administrative files.

### 3.1 Stage 1: Earned income

CPS ASEC microdata (IPUMS, survey years 1980–2023) provides household-level wages, business income, farm income, dividends, interest, rent, and retirement income. Households are ranked by earned income within each survey year using ASEC household weights. Employer-paid benefits (health insurance, retirement contributions) are imputed using year-specific BLS Employer Costs for Employee Compensation (ECEC) ratios applied to wages and salaries. A constant adjustment of

\$1,800 is added to bottom-quintile wages to capture in-kind compensation not captured in the CPS, following Meyer and Sullivan (2012).

### 3.2 Stage 2: Government transfers

National program totals come from administrative sources rather than from self-reported CPS receipts (which substantially underreport in-kind benefits): SSA Trustees Reports for OASI and SSDI; CMS National Health Expenditure tables for Medicare (*net of* beneficiary-paid Part B/D premiums and out-of-pocket cost-sharing); BEA NIPA Table 3.12 for Medicaid, SNAP, and other federal cash and in-kind transfers; OMB Historical Tables for housing assistance; USDA Food and Nutrition Service tables for the National School Lunch Program and WIC. Quintile distribution weights are taken from CBO supplemental Tables 5 and 6, except for OASI receipts, which are re-ranked by earned income from the CPS extract when available.

### 3.3 Stage 3: Tax distribution

Federal taxes — personal income, payroll (OASDI plus Medicare HI), excise, estate and gift, customs, FUTA, railroad retirement, and other retirement — come from CBO Tables 7 and 12 (per-household averages and quintile shares), supplemented with OMB Historical Tables 2.1, 2.4, and 2.5 to break out specific receipt categories. Corporate income tax is excluded from the household burden, since households do not pay it directly.

State and local taxes are distributed from Tax Policy Center / Census State & Local Government Finance national totals using three keys: BLS Consumer Expenditure Survey shares (sales and motor-vehicle taxes), CPS homeownership times household income (property taxes), and CPS wage income (state income taxes). The S&L methodology uses year-specific BLS shares back to 1984 and proxy weights for 1979–1983.

### 3.4 Stage 4: Private transfers and assembly

The Stage 4 script joins the three upstream panels and adds a private-transfer component built from three independent data sources:

- **Component A: Child support and alimony.** CPS variables INCCHILD, INCALIM, and INCALOTH, distributed across quintiles using the Census Child Support Supplement (2018).
- **Component B: Informal cash assistance.** CPS variables INCASIST and INCOTHER, distributed using the normalized bivariate odds ratios from Karen (2023, LIS Working Paper No. 851, Table 2, Model 1).
- **Component C: Charitable giving received by households.** The Giving USA recipient series (1979–2022) is ingested from a CSV with ten recipient categories. Each category is multiplied by a household-directed fraction representing the share of donations that flow through to households as a transfer (Religion 5%, Education 5%, Human Services 50%, Foundations 0%, Health 8%, Public-Society Benefit 5%, International Affairs 25%, Arts and Environment 0%, Individuals 100%). The Giving USA CSV is verified against the *Giving USA 2014 Data Tables* for 1979–2013 and against each subsequent annual highlights publication for 2014–2022.

### 3.5 Stage 5: Ultra-top groups

The CBO supplemental tables provide top-quintile sub-groups (81–90, 91–95, 96–99, and Top 1%). To go above the Top 1%, the script ingests IRS Statistics of Income Table 4.1 Time Series (covers 2001–2022 for the Top 0.1%, 0.01%, and 0.001%) and IRS Top 400 publications (1992–2014, with T41TS-based extrapolations for 2015–2022). IRS Table 3 / 4.3 income-composition files (\*in03etr\*.xls for 2014–2018, \*in43ts\*.xls for 2019–2022) provide labor / capital / pass-through splits at the top of the distribution.

## 4 Findings

### 4.1 The pretax-vs-postfisc gap

Figure 1 plots the top-to-bottom quintile ratio for the two income concepts on a common log scale. The earned-income ratio rises from 24× in 1979 to 96× in 2022 — a fourfold increase that is the basis for most contemporary descriptions of rising inequality. After taxes and transfers, the ratio rises from 2.3× to 4.4× — still upward-trending, but at a much smaller absolute level and at a substantially lower rate of growth. The vertical distance between the two lines is the amount of inequality the standard Census measure misses by ignoring the fiscal system.

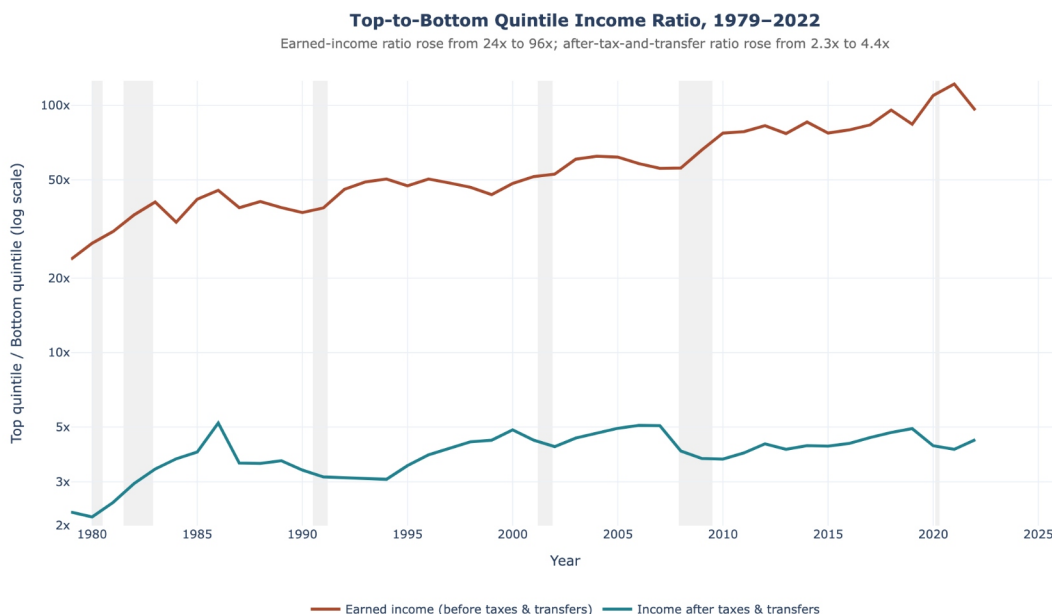


Figure 1: Top-to-bottom quintile income ratio, 1979–2022, comparing earned income to income after taxes and transfers. Both series are reported on a common log scale. The vertical gap between the two lines is the distributional impact of the fiscal system.

## 4.2 The 2022 fiscal cross-section

Figure 2 shows the dollar value of taxes paid, government transfers received, and private transfers received per household per year by quintile in 2022. The bottom quintile is a net fiscal recipient of about \$53k per household per year; the second quintile receives about \$26k net; the middle quintile is roughly neutral (\$1.9k net receipt). The fourth quintile is a net payer of about \$26k, and the top quintile is a net payer of about \$125k. These are dollar amounts at the household level, not percentages of national income.

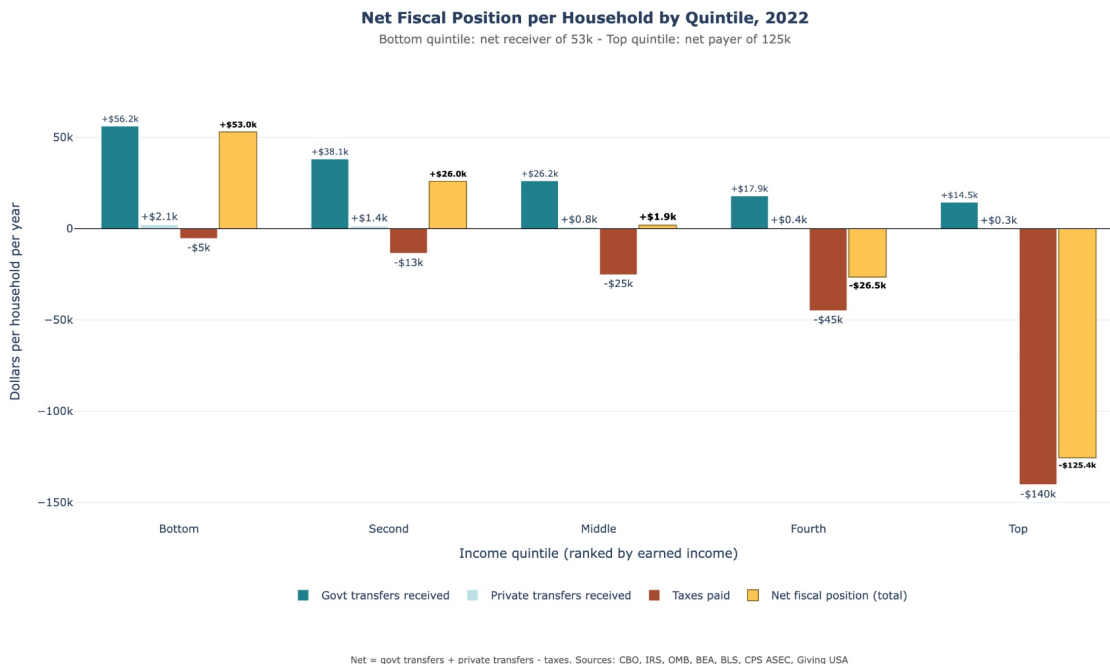


Figure 2: Net fiscal position per household by quintile, 2022. The black bars are the algebraic sum of the other three. Source: panel CSVs in `output/`; underlying data from CBO, IRS, OMB, BEA, BLS CEX, CPS ASEC, and Giving USA.

## 4.3 Choice of deflator: why True CPI rather than CPI-U

Any statement about real income growth depends on the price index used to convert nominal dollars across decades. The default index in most public discussion — the Bureau of Labor Statistics’ CPI-U — is known to overstate the true rate of inflation, and that overstatement compounds over a 43-year horizon into a meaningful distortion of the headline inequality story. The panels in this paper are deflated using the Early and Furth (2021) “True CPI” alternative index. The reasoning is summarized below; the full derivation lives in `src/cpi_bias_adjusted_v3_dbpi.py` and Figure 3.

The academic literature has documented three distinct upward biases in the official CPI-U:

- **Substitution bias** ( $\sim 0.5\text{pp/yr}$ ). The CPI-U is a modified-Laspeyres index that holds expenditure shares roughly constant within a basket update cycle. When relative prices change, consumers substitute toward cheaper alternatives, but the index does not capture this. Estimated at 0.5

percentage points per year by Boskin et al. (1996), Lebow and Rudd (2003), and the BLS Chained CPI-U project. Removing it produces the second curve in Figure 3 (“Chained CPI/PCEPI”), which uses BEA’s PCEPI for 1979–1999 spliced to BLS’s C-CPI-U from 2000 onward.

- **Quality bias.** The CPI assumes a 2022 car, smartphone, or television is the same product as its 1979 counterpart, which it plainly is not. The Boskin Commission estimated this at roughly 0.6pp/yr in the 1990s; Moulton (2018) estimates 0.37pp/yr post-2000 reforms; the bias likely fell further to about 0.30pp/yr after BLS expanded hedonic adjustment to mobile phones in January 2018.
- **New-product bias.** Goods that did not exist in the base period (smartphones, GPS navigation, video streaming, mRNA vaccines) enter the index only after they reach mass adoption, which can be years. Their early-period welfare gains never appear in the price history.

The “True CPI” line in Figure 3 compounds the documented bias schedule on top of the substitution-corrected index, producing a cumulative inflation estimate of 176% from 1979 to 2022 — versus 303% for the official CPI-U over the same window. The 127 percentage point gap is what real-income series in this paper recover relative to series deflated by CPI-U. The gap is concentrated in components where measurement difficulties are best documented: the BLS Disease-Based Price Index (DBPI) for medical care shows roughly 50% cumulative inflation 1999–2017, against 75% for official CPI Medical Care over the same window.

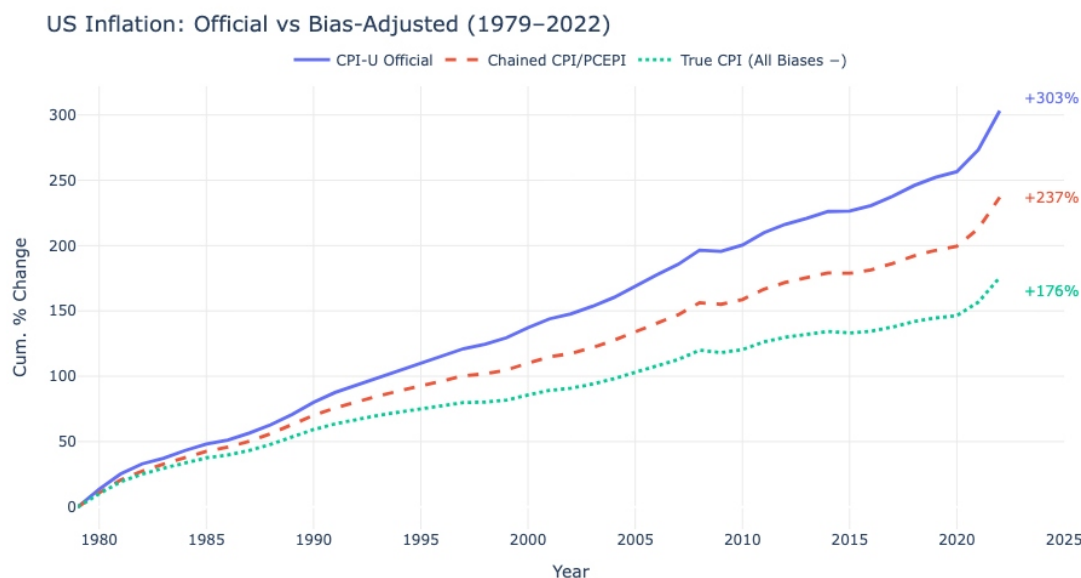


Figure 3: U.S. inflation 1979–2022, three measures. CPI-U is the official BLS series; Chained CPI splices BEA’s PCEPI (1979–1999) to BLS’s C-CPI-U (2000+) and removes substitution bias only; True CPI further deflates the chained series for the documented quality and new-product biases at 0.30–0.60pp/yr depending on the era. The 127pp gap between CPI-U and True CPI by 2022 is the amount by which CPI-U-deflated real series understate long-run real income growth.

Why this matters for the inequality narrative: deflating with CPI-U makes real income growth look smaller for everyone, but the effect is not distributionally neutral. The bottom quintile’s real

after-tax-and-transfer income grew by about 129% over 1979–2022 under True CPI; under CPI-U the same nominal change produces only about 33% real growth. The qualitative story shifts from “every quintile gained substantially in real terms” to “the bottom quintile barely moved” — not because the dollar flows changed, but because of which deflator was used. Choosing a deflator known to overstate inflation by roughly 0.9pp/yr is therefore not a neutral modeling choice; it is itself a determinant of the narrative.

This paper uses True CPI throughout for two reasons: (a) it more accurately reflects the basket of goods households actually consumed each year, given quality and product-mix changes that the official index misses; and (b) it is the index used by Gramm, Ekelund, and Early in *The Myth of American Inequality*, the closest published precedent for this style of analysis. Sensitivity to deflator choice can be reproduced by re-running the charts against the `CPI_U_idx` column of `cpi_adjusted_v3_dbpi_1979_2022.csv`, which is included in the same output file.

#### 4.4 Real income growth, 1979–2022

Figure 5 reports real income after taxes and transfers by quintile from 1979 to 2022, deflated using the Early and Furth (2021) “True CPI” alternative price index. After 43 years, the real after-tax-and-transfer income of the bottom quintile has risen by 129%, the middle quintile by 201%, and the top quintile by 349%. Every quintile has experienced positive real growth in income that is actually available for consumption.

The contrast with earned income is sharp: real earned income for the bottom quintile has *fallen* by 48% over the same period (Figure 4), even as their real after-tax-and-transfer income rose. The fiscal system absorbs the lost earnings entirely and turns them into a small net gain.

#### 4.5 A note on quintile ordering in the early years

A careful reader will notice that in Figure 5 the Bottom and Second quintile lines nearly overlap (and slightly cross) for the first few years of the panel. This is not a measurement error but a real feature of the late-1970s post-fisc distribution. In 1979 the per-household IAT for the bottom quintile is \$9,029, slightly above the second quintile’s \$8,637. The full numerical ordering violations occur in 1979–1983 and briefly in 1985.

Two factors drive this. First, the bottom quintile in the late 1970s was disproportionately retired-elderly households living on Social Security: their earned income was very low (about \$2,800 in 1979), but they received large OASI benefits (about \$8,100 per household) and paid almost no income tax (retirement income was largely untaxed in that era). Second, the second quintile was disproportionately working households at the bottom of the wage distribution: they earned roughly five times more in the market (\$13,400) but paid roughly four times more in taxes (\$7,800), and received only \$3,000 in transfers because they were too young for Social Security and not poor enough for cash welfare.

The ordering normalizes after about 1985 for three reasons: the Earned Income Tax Credit became more generous (raising second-quintile post-tax income), AFDC and other cash welfare programs were tightened (lowering bottom-quintile transfers), and the elderly-retired share of the bottom quintile gradually fell as more two-earner working households entered the bottom of the earned-income distribution. Strictly increasing post-fisc income across quintiles is therefore not a property the panel imposes; it is something that emerged endogenously over time as the demographic

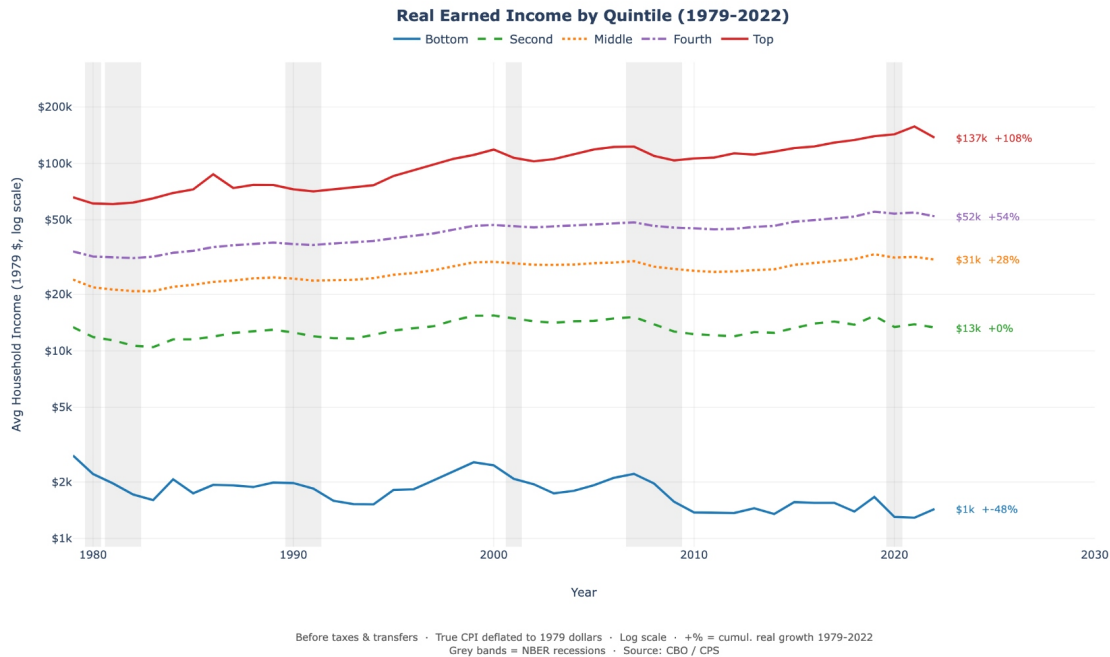


Figure 4: Real earned income by quintile, 1979–2022. Deflated using Early and Furth (2021) True CPI. Real earned income at the bottom of the distribution has *fallen* since 1979, while the top has more than doubled.

composition of the bottom quintile shifted.

## 4.6 Effective tax rates

Figure 6 plots a signed effective tax rate,  $(T - G - P)/E$ , by quintile from 1979 to 2022. The bottom quintile's signed rate has fallen sharply over time and is now strongly negative (the household receives far more in transfers than it pays in taxes). The middle quintile's rate has fallen from roughly 30% positive in 1979 to roughly zero in 2022. The top quintile pays approximately 33% of earned income on net.

## 4.7 The very top of the distribution

Figure 7 reports real income after taxes and transfers for groups above the top quintile: the Top 1%, Top 0.1%, Top 0.01%, Top 0.001%, and the Forbes 400 (years vary by data availability). The signed effective tax rate for the Forbes 400 in 2022 is 26.1%, lower than the Top 1% rate of 31.6%. The differential reflects the larger share of capital gains in AGI at the very top (taxed at preferential rates) and the cap on payroll taxes.



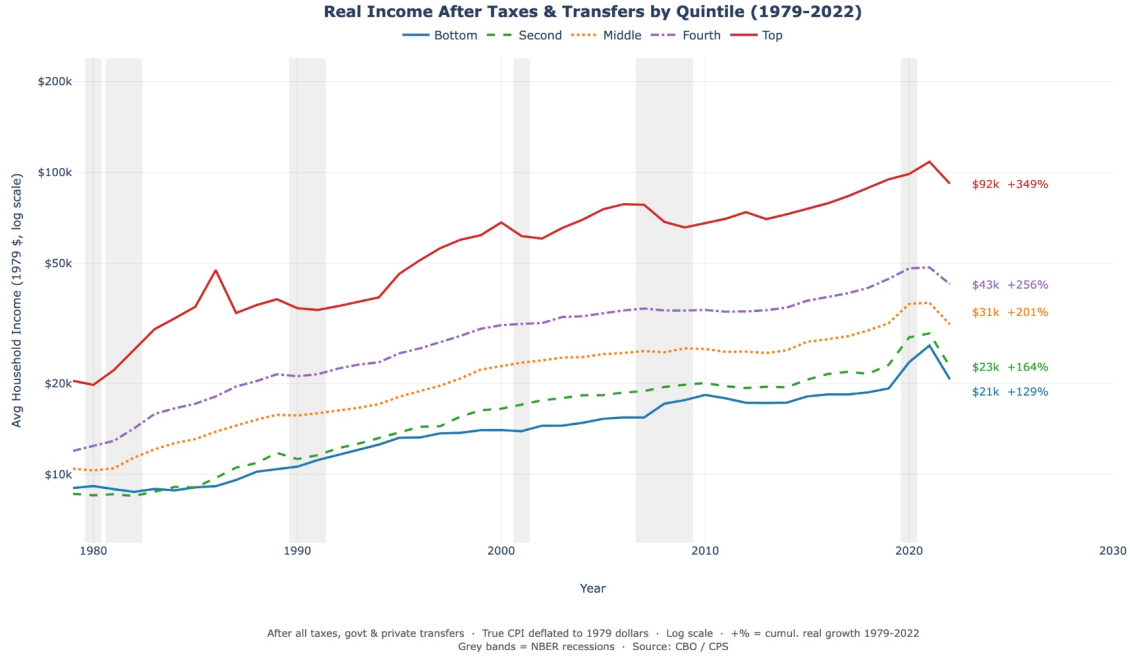


Figure 5: Real income after taxes and transfers by quintile, 1979–2022. Once taxes and transfers are accounted for, every quintile has experienced positive real growth.

## 5 Discussion

The size of the discrepancy between the Census “money income” measure and a complete post-fisc measure is large in both cross-section and time series. In 2022, the bottom quintile receives \$56k of government transfers per household per year on average, of which a large share is in-kind — not counted by the Census construct. The top quintile pays \$140k of taxes per household per year, none of which is netted out of the Census construct. The combined effect is to shrink the top-to-bottom quintile gap by an order of magnitude ( $96\times$  becomes  $4.4\times$ ).

This does not imply that inequality is fictional, nor that distributional concerns should be set aside: the underlying *earned-income* ratio of  $96\times$  is a real reflection of differences in market productivity. But it does mean that any claim about the trajectory or level of inequality needs to specify which income concept is being measured. The choice of concept is consequential: in 1979 the bottom-quintile household took home about 22% more in transfers than it paid in taxes; in 2022 it takes home about  $14\times$  more.

### 5.1 Limitations and open questions

Several judgment calls are baked into the panel and should be flagged.

- **Medicare valuation.** The panel reports Medicare *net of* beneficiary-paid Part B/D premiums and cost-sharing. An alternative is to value Medicare at its insurance-equivalent value, which is substantially lower than the gross outlay, particularly at the top end of the senior distribution.

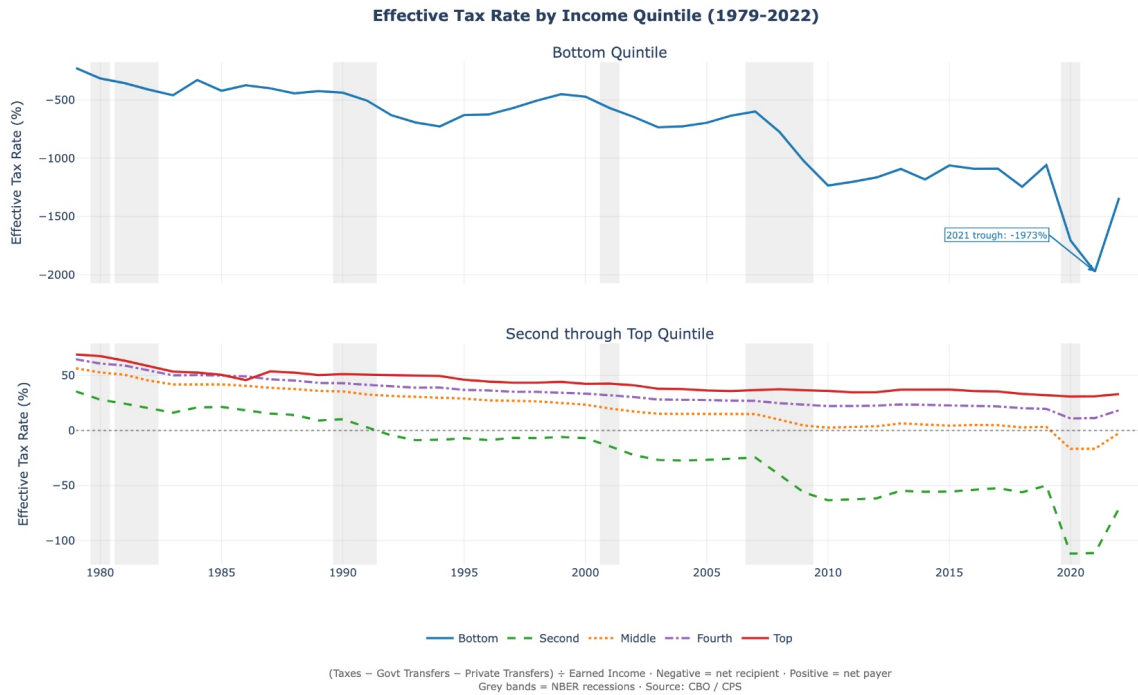


Figure 6: Signed effective tax rate by income quintile, 1979–2022. The numerator is taxes paid minus all transfers received; the denominator is earned income.

- **Tax incidence on corporate income.** We exclude corporate income tax from the household burden. The tax-incidence literature is not unanimous about whether the burden falls primarily on capital or on labor; including it shifts both the level and the slope of Figure 6.
- **Private transfer methodology.** The Component C approach (multiplying each Giving USA category by a fixed household-directed fraction) gives an estimate, not a measurement. The fractions for Religion (5%), Health (8%), and International Affairs (25%) are defensible as central estimates but have wide bounds. Sensitivity analysis using upper and lower bounds for those fractions is possible from the panel.
- **Census deflator.** We use the Early and Furth (2021) True CPI index. The standard CPI-U over-states inflation by roughly 0.5–0.8 percentage points per year; using the CPI-U instead of the True CPI would lower the measured real growth of every quintile.

## 6 Proposed extensions

This work suggests several directions for additional analysis that the existing panel infrastructure can support without substantial new data collection.

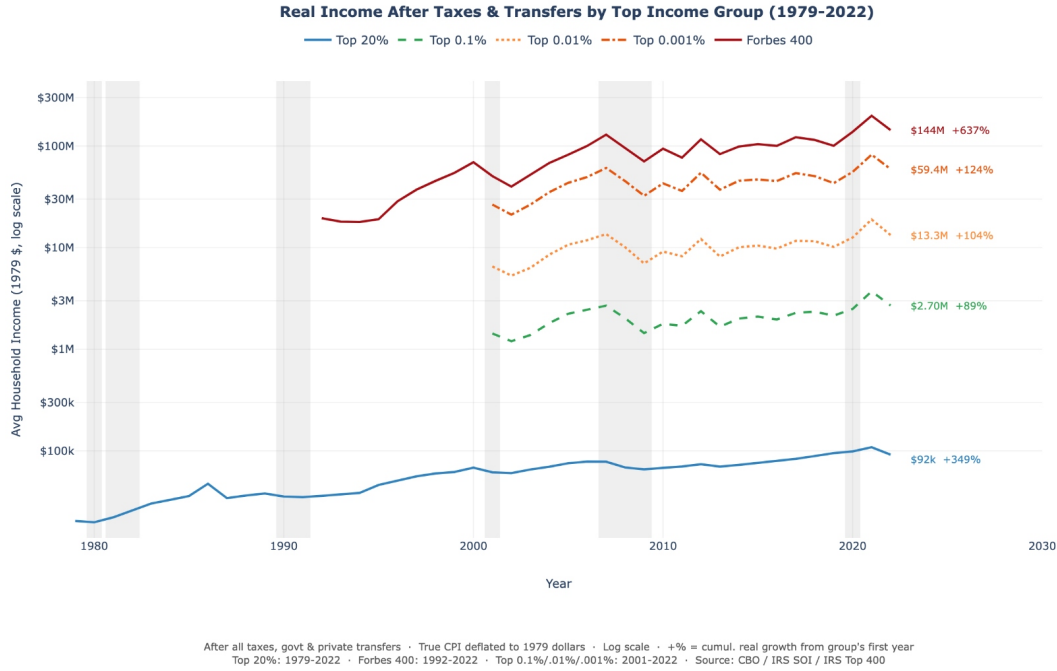


Figure 7: Real income after taxes and transfers for ultra-top groups, 1979–2022 (years vary by data availability).

## 6.1 Decomposition of the 1979–2022 IAT change

A Shapley-style decomposition of the change in real after-tax-and-transfer income for each quintile, allocating the total to four components: change in earned income, change in transfers received, change in taxes paid, and change in the implicit price index. This would clarify how much of each quintile’s gain over four decades is attributable to market earnings versus the fiscal system.

## 6.2 Replication using alternative tax-incidence assumptions

Re-run the panel under alternative assumptions for who bears the corporate income tax burden (100% capital, 50/50 capital–labor, 100% labor) and report the resulting effective rates. The Auerbach (2018) and Suarez Serrato–Zidar (2016) literature suggests this is a meaningful sensitivity.

## 6.3 Real consumption analogues

The companion BLS Consumer Expenditure Survey panel already feeds the S&L sales-tax distribution; the same data can be used to construct a household real-consumption series by quintile, providing a direct test of whether consumption inequality has tracked income inequality (Krueger and Perri 2006 suggest it has not).

## 6.4 Cross-cohort mobility

The panel as constructed treats each year’s quintile membership as independent. The Panel Study of Income Dynamics (PSID) and IRS Statistics of Income longitudinal samples allow the same income concepts to be computed across cohorts, permitting an explicit test of upward and downward mobility in post-fisc terms over the past four decades.

## 7 Reproducibility

All seven Python scripts and the underlying CSV outputs are available at [github.com/Patrickkp1/tax-transfer-inequality](https://github.com/Patrickkp1/tax-transfer-inequality). Each script is self-contained, walks up its own directory tree to find the `data/raw/` folder, and can be re-run end-to-end on any UNIX-like machine with Python 3.10+, pandas, numpy, openpyxl, and plotly. Total end-to-end runtime against a full CPS extract is approximately 30 minutes. Each panel CSV is byte-identical across runs given the same inputs.

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